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Topic :

The Impact of Social Media Usage on Academic Performance and Productivity Behaviours Among Students

Research Methodology — Survey-Based Study

Target Population: University Students

Age Group: 18–24 Years

Research Design: Cross-Sectional Questionnaire Survey

Sample Size (n): 50 Students

Data Collection Method: Google Form Questionnaire

Background:

Social media has become deeply embedded in the daily routines of university students, yet its precise relationship with academic performance and productivity remains contested.

Methods: A cross-sectional questionnaire survey ($n = 50$) of students aged 18–24 assessed social media usage patterns, focus reduction, procrastination, coursework distraction, time management difficulty, notification disruption, multitasking behaviour, and self-reported academic and productivity impact. Six pre-registered hypotheses were evaluated using Spearman rank correlations, Mann–Whitney U tests, Kruskal–Wallis tests, and OLS multiple regression. A composite Distraction Score was computed as the mean of six behavioural items.

Results: Social media use during study sessions was significantly associated with focus reduction ($\rho = 0.384$, $p = 0.006$) and elevated composite distraction ($\rho = 0.374$, $p = 0.007$). Procrastination was the strongest predictor of academic impact ($\rho = 0.430$, $p = 0.002$), with the composite distraction score showing a strong correlation with academic impact ($\rho = 0.506$, $p < 0.001$). OLS regression revealed that procrastination, focus reduction, and study-session usage jointly explained 20.1% of variance in academic impact ($F(3,46) = 3.849$, $p = 0.015$). No significant gender differences in distraction were observed. Platform type did not significantly predict academic impact.

Conclusion: The impact of social media on academic performance operates primarily through behavioural mechanisms — in-session usage, focus disruption, and procrastination — rather than through total daily exposure alone. Interventions targeting these specific behaviours may be more effective than blanket usage restrictions

Introduction

The proliferation of social media platforms — Instagram, YouTube, TikTok, WhatsApp, Snapchat, and their derivatives — has fundamentally altered the way young adults manage their time, attention, and interpersonal relationships. Among university students, social media usage is now nearly universal, with daily engagement times frequently exceeding three hours (Anderson and Jiang, 2018). This temporal investment raises an important empirical question: to what extent does social media use undermine the academic productivity and performance of students who are simultaneously navigating the demands of higher education?

Over the past decade, higher educational institutions globally have witnessed a seismic shift in student behaviour, particularly concerning the use of digital technologies. Where previous generations of students faced relatively discrete boundaries between leisure and study, contemporary undergraduates carry with them, at virtually all times, devices that provide instant access to a rich ecosystem of social, informational, and entertainment content. The smartphone has effectively collapsed the boundary between study space and social space, and social media platforms have been the primary beneficiaries of this collapse.

Globally, the average daily social media usage among young adults aged 18–24 has risen consistently year-on-year, reaching an estimated 3.1 hours per day by 2023 (Statista, 2024). Within Indian universities specifically — the context of the present study — the landscape is shaped by a unique convergence of factors: a rapidly expanding smartphone penetration rate, affordable mobile data plans following the entry of Jio in 2016, and a youthful demographic that is among the most digitally connected in the world. India is now one of the largest markets for Instagram, YouTube, and WhatsApp, with the 18–24 age cohort representing the most active user segment (IAMAI, 2023).

The academic consequences of this pervasive digital engagement are, however, far from settled in the scholarly literature. A substantial body of work documents significant negative associations between social media use and academic outcomes, including reduced grade point averages (GPAs), poorer assignment quality, missed deadlines, and lower overall academic engagement (Kirschner and Karpinski, 2010; Paul et al., 2012; Junco, 2012). Yet competing evidence suggests that social media may, under certain conditions, support collaborative learning, peer networking, resource sharing, and academic motivation (Dabbagh and Kitsantas, 2012; Pasek et al., 2009). This heterogeneity of findings is not merely a matter of conflicting evidence — it reflects genuine complexity in the mechanisms through which social media interacts with academic life.

A central limitation of much prior research is its reliance on total daily screen time or platform membership as the primary measure of social media exposure. These blunt instruments fail to capture the nuanced ways in which students actually engage with social media in academically relevant contexts. A student who spends four hours on YouTube watching educational content in the evening occupies a fundamentally different cognitive and motivational position than one who checks Instagram every ten minutes during a lecture or problem-solving session. The context, frequency, and nature of usage — rather than aggregate duration — may be the critical determinants of academic consequences.

This insight has motivated a shift in the theoretical framing of social media research, away from simple dose-response models toward a more nuanced examination of behavioural mediators. Cognitive Load Theory (Sweller, 1988) provides a foundational framework: working memory has finite capacity, and social media notifications and the habitual checking behaviours they trigger impose extraneous cognitive load that competes with the germane processing required for deep learning. Attention Fragmentation Theory (Ophir et al., 2009) extends this by documenting the cumulative attentional consequences of chronic multitasking — a behaviour ubiquitous among heavy social media users. The reward architecture of social media platforms themselves — specifically the variable reinforcement schedules of likes, comments, and notifications — creates the psychological conditions for compulsive checking and, by extension, academic procrastination (Alter, 2017; Steel, 2007).

The present study is situated within this evolving theoretical landscape. Rather than examining total daily social media hours as the primary independent variable, it constructs a composite Distraction Score that aggregates six behavioural mediators: usage during study sessions, focus reduction, coursework distraction, time management difficulty, multitasking behaviour, and study delay. This composite is intended to capture the multi-dimensional nature of social media's disruptive academic influence — moving beyond what students use and for how long, toward a richer characterisation of how they use it.

Beyond individual-level behavioural mechanisms, the study also examines structural factors that have received relatively limited empirical attention in the Indian higher education context: gender differences in social media distraction, and the differential academic consequences of specific platforms. Gender differences in social media usage patterns are well-documented, with females showing higher rates of engagement with visually-oriented platforms such as Instagram and males showing stronger preferences for YouTube and gaming platforms (Statista, 2024). Whether these usage differences translate into differential academic consequences is an empirical question with meaningful implications for intervention design. Similarly, the platform-specific nature of social media — with TikTok's algorithm-driven short-form video content, Instagram's visual reward architecture, and YouTube's longer-form engagement patterns — may produce qualitatively different attentional demands that modulate academic disruption.

Six pre-registered hypotheses are tested in the current study, addressing: the relationship between daily social media hours and academic impact (H1); the association between in-session usage and focus reduction (H2); the link between daily hours and procrastination (H3); the paradoxical relationship between usage and perceived productivity impact (H4); gender differences in composite distraction (H5); and platform-type effects on academic impact (H6). Statistical analysis employs Spearman rank correlation, Mann–Whitney U tests, Kruskal–Wallis tests, and OLS multiple regression, with non-parametric methods selected on the basis of confirmed non-normality via Shapiro–Wilk tests.

The study is structured as follows. Section 2 presents a comprehensive review of the theoretical and empirical literature, organised around the key mechanisms of interest: social media and academic performance, procrastination as a mediating mechanism, attention fragmentation and focus reduction, and platform-specific and gender effects. Section 3 articulates the six study hypotheses. Section 4 describes the research design, sampling strategy, survey instrument, variable operationalisation, and data analysis plan. Section 5 presents the results across descriptive statistics, hypothesis tests, extended correlation analysis, regression models, and group comparisons. Section 6 discusses the findings in theoretical and practical context. Sections 7 and 8 address policy implications and conclusions

Literature review

2.1 Social Media and Academic Performance: An Overview

The academic study of social media's impact on student performance emerged rapidly in the late 2000s, driven by the explosive growth of Facebook as a campus phenomenon. Early research by Kirschner and Karpinski (2010) established a foundational finding: Facebook users reported lower GPAs and spent fewer hours per week studying than non-users, even after controlling for other variables. This initial contribution sparked a proliferation of studies across diverse national contexts, with broadly convergent results indicating a negative association between social media engagement and academic performance, though with considerable variation in effect sizes and methodological approaches.

Junco (2012) advanced the field by moving beyond mere platform membership to examine the specific nature of social media behaviour. His survey of 1,839 college students demonstrated that the activity of status-updating and checking — rather than total time spent on Facebook — was most predictive of negative academic outcomes. This distinction between passive consumption and active, interruptive checking has since become a recurring theme in the literature, with multiple researchers confirming that the manner and context of use are more consequential than duration alone (van Deursen and van Dijk, 2014)

Not all evidence points in a negative direction. Pasek, More, and Hargittai (2009) found no significant relationship between Facebook use and GPA after appropriate statistical controls, cautioning against the assumption of a simple causal pathway. Dabbagh and Kitsantas (2012) proposed a theoretical framework in which social media, when used intentionally for academic purposes — group study coordination, resource sharing, collaborative problem-solving — could function as a personal learning environment that enhances rather than undermines academic engagement. This perspective has found empirical support in studies of specific academic uses of platforms like Twitter and LinkedIn (Junco, Heiberger, and Loken, 2011).

The theoretical foundation for understanding negative academic effects has increasingly drawn on Cognitive Load Theory (Sweller, 1988). This framework posits that human working memory has a strictly finite capacity for simultaneous information processing. Academic tasks — reading complex texts, solving mathematical problems, writing analytical essays — require sustained germane cognitive load, the mental effort directed toward schema formation and deep learning. Social media notifications and the habitual checking behaviour they trigger impose extraneous cognitive load that competes with this germane processing, effectively crowding out the attentional resources required for effective learning. Rosen, Carrier, and Cheever (2013) provided compelling experimental evidence for this mechanism, demonstrating that brief Facebook-related interruptions during study sessions significantly reduced comprehension and recall of academic material compared to uninterrupted study conditions

2.2 Procrastination as a Central Mediating Mechanism

Procrastination — formally defined as the voluntary delay of intended tasks despite knowledge of expected negative consequences (Steel, 2007) — has emerged as arguably the most important psychological mechanism linking social media use to poor academic outcomes. Steel's landmark meta-analysis of 691 correlational studies confirmed procrastination as the most prevalent form of academic self-regulation failure, with estimates suggesting that 70–95% of university students engage in some form of academic procrastination (Rozental and Carlbring, 2014).

Social media platforms are specifically designed to exploit the same reward-seeking tendencies that underpin procrastination. The variable reinforcement schedules embedded in social media interaction — the unpredictable delivery of likes, comments, shares, and notifications — create powerful approach motivations analogous to those observed in slot machine behaviour (Alter, 2017). When confronted with a cognitively demanding academic task that offers delayed and uncertain rewards, students face a powerful temptation to switch their attention toward social media, which offers immediate, frequent, and socially validated rewards. This temporal discounting of academic rewards in favour of social media gratification is a hallmark of procrastinatory behaviour.

Andreassen et al. (2016), in a large-scale national Norwegian survey, found significant positive associations between addictive social media use and academic procrastination, mediated partly through narcissistic personality traits and poor self-esteem. More recently, Kim (2020) demonstrated in a longitudinal study that procrastination fully mediated the relationship between social media use and academic performance decline, confirming its role as a critical pathway rather than a mere correlate. The current study contributes to this body of evidence by examining procrastination as both an individual predictor and a component of a broader composite distraction measure within an Indian student sample.

2.3 Attention Fragmentation and Focus Reduction

Complementary to the procrastination pathway is the mechanism of attention fragmentation, elaborated most influentially by Ophir, Nass, and Wagner (2009) in their study of media multitaskers. Their experimental findings demonstrated that individuals who habitually engage in media multitasking — simultaneously managing multiple streams of digital information — show reduced ability to filter irrelevant stimuli, maintain sustained attention, and switch efficiently between tasks relative to light media multitaskers. These attentional deficits represent a form of cognitive cost imposed by the chronic interruption patterns characteristic of heavy social media use.

For university students, these attentional consequences translate into practical academic difficulties: reduced comprehension of lecture material when devices are present, slower reading speeds during independent study, greater difficulty sustaining concentration through complex problem-solving, and impaired recall of studied material. The composite Distraction Score constructed in the present study — aggregating usage during study, focus reduction, coursework distraction, time management difficulty, multitasking, and study delay — is designed to operationalise these multi-faceted attentional consequences within a single summary measure.

Research Hypotheses

Based on the theoretical frameworks and empirical evidence reviewed above, six hypotheses are tested at the conventional significance threshold of $\alpha = 0.05$:

Hypothesis	Statement
H1	Higher daily social media hours are positively associated with greater self-reported academic impact.
H2	Social media use during study sessions is positively associated with focus reduction.
H3	Higher daily social media hours are positively associated with procrastination frequency.
H4	Higher daily social media hours are negatively associated with perceived productivity impact.
H5	Significant gender differences exist in composite distraction scores attributable to social media use.
H6	Platform type (Instagram vs. YouTube vs. others) differs in its association with academic impact.

Methodology

4.1 Research Design and Sampling

A cross-sectional survey design was employed. The target population comprised undergraduate students aged 18–24 at Indian universities. Convenience sampling was used to recruit respondents via an online Google Forms questionnaire distributed through academic networks. A total of 50 complete responses were obtained. The sample consisted of 39 females (78%) and 11 males (22%), all within the 18–24 age group. While convenience sampling limits strict probabilistic generalisation, the sample size and homogeneity of the target demographic provide a meaningful foundation for hypothesis testing and theory-building.

4.2 Survey Instrument

The questionnaire comprised 19 items across two sections. Section 1 collected demographic information (age group, gender). Section 2 measured social media behaviour and its consequences through 17 items covering daily usage hours, frequency of use during study, focus reduction, procrastination, coursework distraction, time management difficulty, notification disruption, multitasking, study delay, usage-limitation attempts, platform preference, academic performance impact frequency, productivity when avoiding social media, break frequency, and overall academic productivity impact.

4.3 Variable Operationalisation

Ordinal response categories were encoded numerically for analysis. Frequency scales (Rarely / Sometimes / Often / Always) were coded 1–4; impact scales (Not at all / Slightly / Moderately / Significantly) were coded 0–3. A composite Distraction Score was computed as the mean of six behavioural items: usage during study (Q4), focus reduction (Q5), coursework distraction (Q8), time management difficulty (Q9), multitasking (Q11), and study delay (Q13). Higher scores indicate greater distraction.

4.4 Ethical Considerations All responses were collected anonymously. Participation was voluntary, with informed consent obtained through the survey preamble. No personally identifiable information was collected.

4.5 Data Analysis Plan

Research Methods Department All analyses were conducted in Python using scipy, numpy, and pandas. The analysis plan comprised: (1) descriptive statistics; (2) Shapiro–Wilk normality tests; (3) Spearman rank correlation analysis; (4) Mann–Whitney U test for gender comparisons; (5) Kruskal–Wallis test for platform comparisons; and (6) OLS multiple regression (three models).

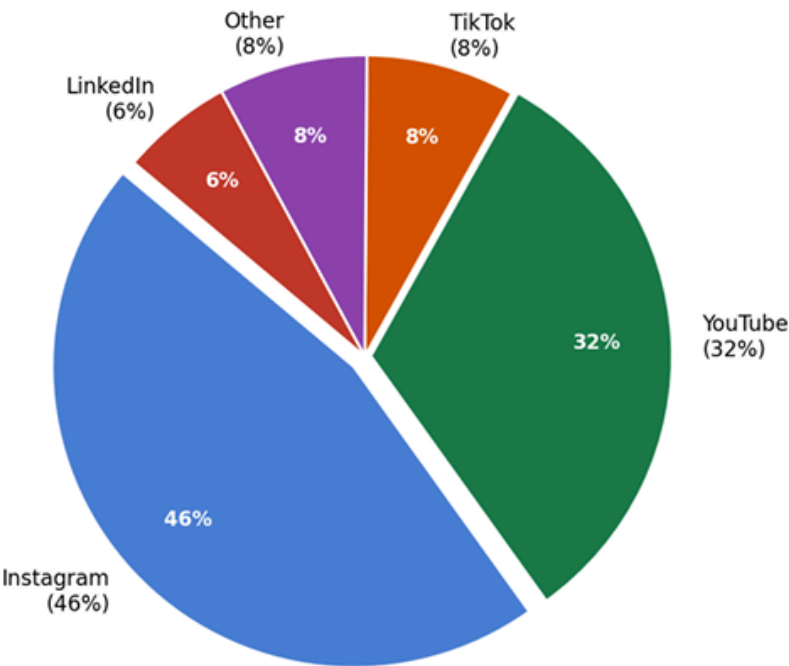
Analysis

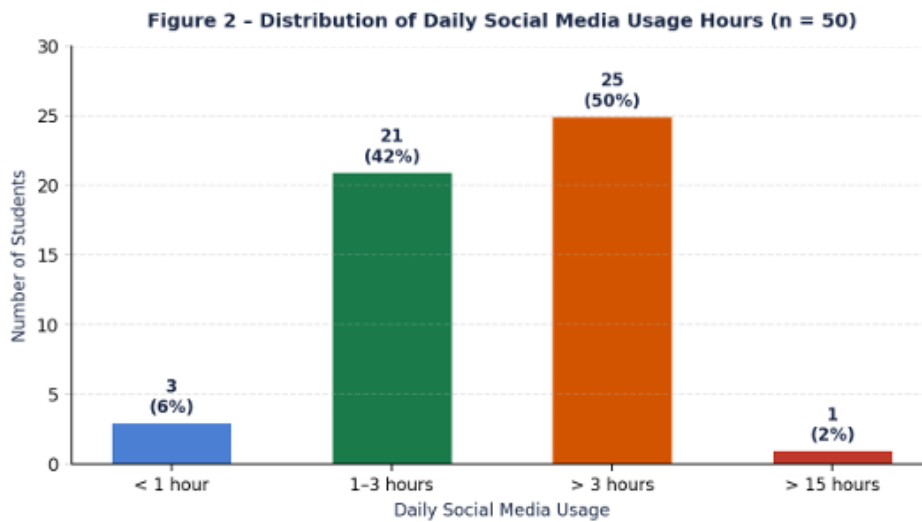
5.1 Sample Profile

Research Methods Department The sample (n = 50) comprised exclusively students in the 18–24 age group. Gender distribution was 39 females (78.0%) and 11 males (22.0%). The majority of respondents reported using social media for more than 3 hours per day (n = 25, 50%), followed by 1–3 hours (n = 21, 42%). Instagram was the most used platform (n = 23, 46%), followed by YouTube (n = 16, 32%).

Variable	Category	n	%	Cumulative %
Gender	Female	39	78.0%	78.0%
	Male	11	22.0%	100.0%
SM Hours/Day	> 3 hours	25	50.0%	50.0%
	1–3 hours	21	42.0%	92.0%
	< 1 hour	3	6.0%	98.0%
	> 15 hours	1	2.0%	100.0%
Platform	Instagram	23	46.0%	46.0%
	YouTube	16	32.0%	78.0%
	TikTok	4	8.0%	86.0%
	Other	4	8.0%	94.0%
	LinkedIn	3	6.0%	100.0%

Figure 1 - Most-Used Social Media Platform (n = 50)



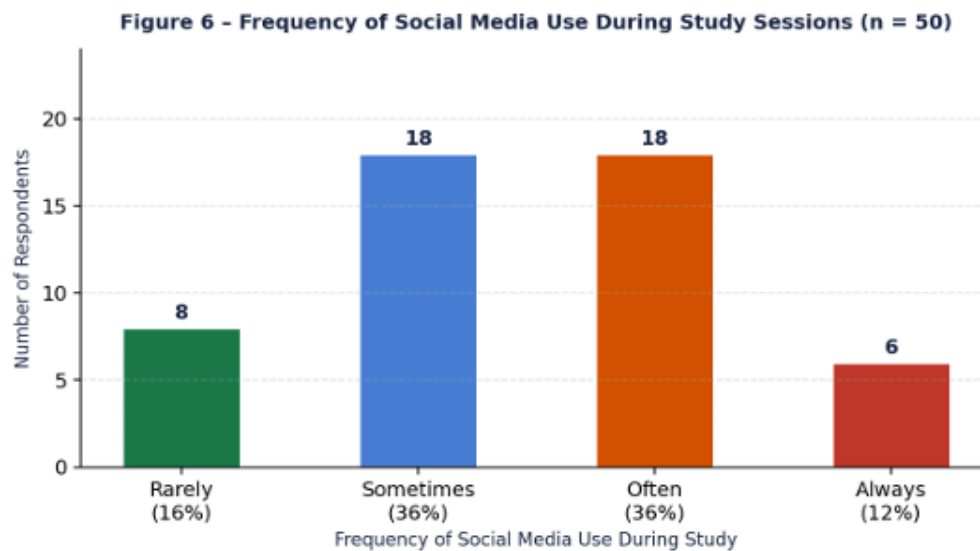


5.2 Descriptive Statistics

Table 2 presents descriptive statistics for all key numeric variables. Social media usage during study sessions showed a mean of 2.44 (SD = 0.907), indicating that the modal response was between "Sometimes" and "Often". Focus reduction was moderate on average (M = 1.70, SD = 0.839), and procrastination was relatively prevalent (M = 2.40, SD = 0.728). The composite Distraction Score had a mean of 1.85 (SD = 0.613). Shapiro–Wilk tests confirmed that all key variables significantly departed from normality ($p < 0.001$), justifying non-parametric inferential tests throughout.

Variable	N	Mean	Median	SD	Min	Max	Skewness
SM Hours	50	2.48	3	0.646	1	4	-0.383
Usage During Study	50	2.44	2	0.907	1	4	0.014
Focus Reduction	50	1.70	2	0.839	0	3	0.188
Procrastination	50	2.40	2	0.728	1	4	-0.128
Coursework Distraction	50	1.38	1	0.753	0	3	0.125
Time Mgmt Difficulty	50	1.34	1	0.798	0	3	0.048
Academic Impact	50	1.42	1	0.810	0	3	0.261
Productivity Impact	50	2.00	2	0.833	0	3	-0.214
Distraction Score	50	1.85	1.83	0.613	0.50	3.33	0.142

Table 2 – Descriptive Statistics: Key Study Variables

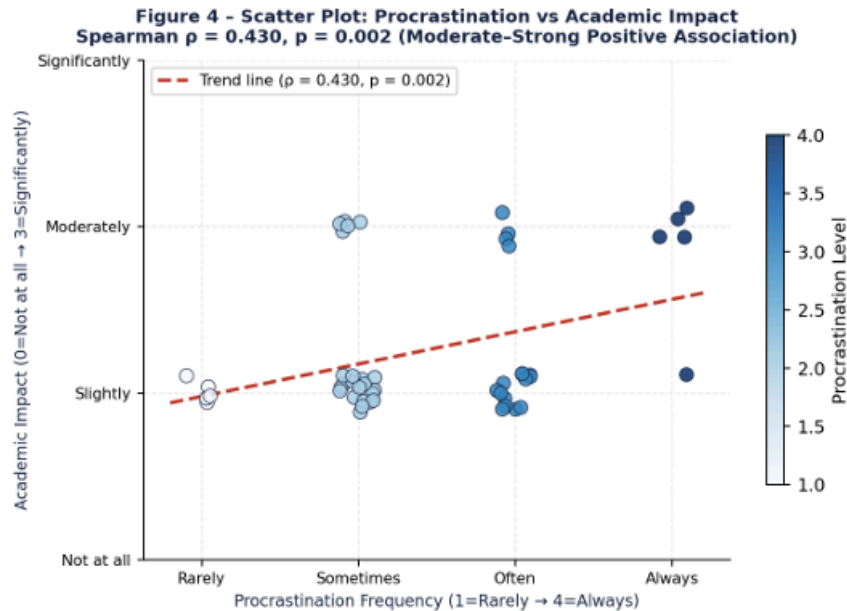


5.3 Hypothesis Tests — Spearman Correlation Analysis

Given confirmed non-normality in all variables, Spearman rank correlation (ρ) was used for all bivariate inferential tests. Table 3 presents the full results for all six hypotheses.

H	Statement	Statistic	p-value	Effect Size	Decision
H1	SM hours → Academic impact	$\rho = 0.028$	0.846	Negligible	× Not Supported
H2	SM use during study → Focus reduction	$\rho = 0.384$	0.006**	Moderate	✓ Supported
H3	SM hours → Procrastination	$\rho = 0.255$	0.074	Small-Mod.	~ Marginal
H4	SM hours → Lower productivity impact	$\rho = -0.284$	0.046*	Small-Mod.	✓ Supported
H5	Gender → Distraction Score	$U = 248.5$	0.430	$r_b = -0.159$	× Not Supported
H6	Platform → Academic impact	$H = 3.932$	0.415	η^2 not sig.	× Not Supported

Table 3 – Hypothesis Testing Results (n = 50, $\alpha = 0.05$). * $p < 0.05$; ** $p < 0.01$. ✓ = Supported; × = Not Supported; ~ = Marginal.



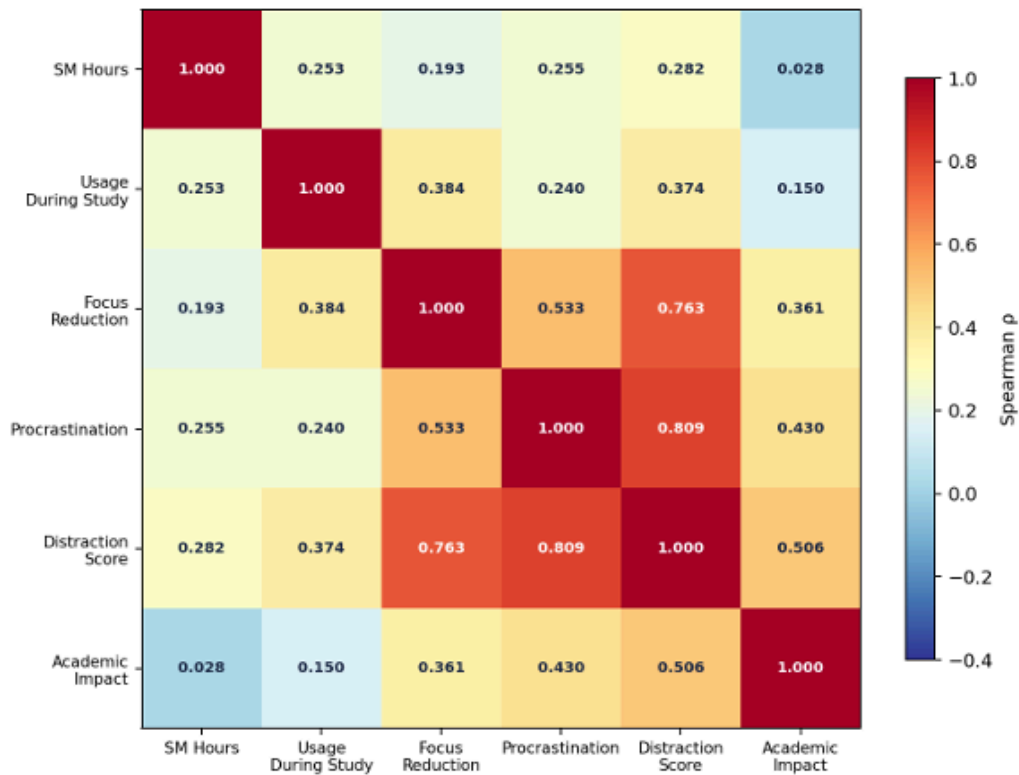
5.4 Extended Correlation Analysis

Beyond the primary hypotheses, Table 4 presents the full Spearman correlation matrix for all key behavioural variables. The composite Distraction Score demonstrates exceptional predictive power, correlating strongly with procrastination ($\rho = 0.809$) and focus reduction ($\rho = 0.763$), and also with overall academic impact ($\rho = 0.506$).

Variable Pair	ρ	p-value	Interpretation
Focus Reduction × Procrastination	0.533	< 0.001***	Strong: focus loss drives procrastination
Focus Reduction × Academic Impact	0.361	0.010*	Moderate: concentration losses harm grades
Procrastination × Academic Impact	0.430	0.002**	Moderate-strong: delays hurt performance
Distraction Score × Focus Reduction	0.763	< 0.001***	Very strong: distraction core dimension
Distraction Score × Procrastination	0.809	< 0.001***	Very strong: distraction drives delays
Distraction Score × Academic Impact	0.506	< 0.001***	Strong: distraction predicts impact
Usage During Study × Distraction Score	0.374	0.007**	Moderate: in-session use inflates distraction
SM Hours × Productivity Impact	-0.284	0.046*	Negative: heavy users rationalise usage
SM Hours × Focus Reduction	0.193	0.179 ns	Non-significant: hours do not predict focus
SM Hours × Academic Impact	0.028	0.846 ns	Non-significant: hours not linked to impact

Table 4 – Extended Spearman Correlation Matrix ($n = 50$). ns = not significant; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

**Figure 5 - Spearman Rank Correlation Matrix
Key Study Variables (n = 50)**



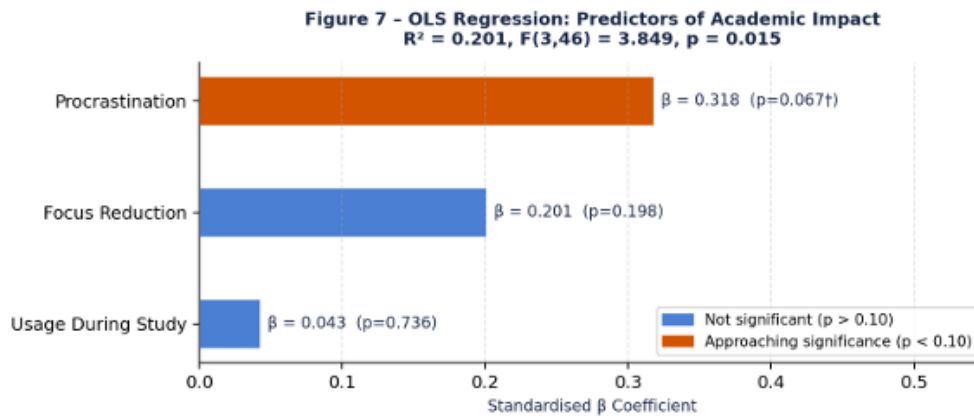
5.5 OLS Multiple Regression Analysis

5.5.1 Model 1: Predicting Academic Impact

The first model regressed self-reported academic impact on usage during study, focus reduction, and procrastination frequency. The model was statistically significant ($F(3,46) = 3.849$, $p = 0.015$) and explained 20.1% of variance in academic impact ($R^2 = 0.201$, Adjusted $R^2 = 0.149$). Procrastination emerged as the strongest individual predictor ($\beta = 0.318$, $t = 1.875$, $p = 0.067$), approaching significance.

Predictor	β	SE	t-stat	p-value	Interpretation
Intercept	0.210	0.426	0.492	0.625	—
Usage During Study	0.043	0.127	0.339	0.736	Non-significant
Focus Reduction	0.201	0.154	1.305	0.198	Positive trend
Procrastination	0.318	0.170	1.875	0.067†	Approaching sig.

Table 5 – OLS Regression Model 1: Predicting Academic Impact ($R^2 = 0.201$, $F(3,46) = 3.849$, $p = 0.015^$)*



5.5.2 Model 2: Predicting Composite Distraction Score

The second model regressed the Distraction Score on SM hours and usage during study sessions. The model was significant ($F(2,47) = 4.260$, $p = 0.020$) and explained 15.3% of variance ($R^2 = 0.153$). Usage during study was the only significant predictor ($\beta = 0.245$, $t = 2.652$, $p = 0.011$), confirming that in-session behaviour — not total daily hours — drives composite distraction. Predictor Intercept SM Hours Usage During Study β 1.025 0.090 SE 0.365 t-stat 2.806 p-value

Predictor	β	SE	t-stat	p-value	Interpretation
Intercept	1.025	0.365	2.806	0.007**	Baseline distraction
SM Hours	0.090	0.130	0.695	0.490	Non-significant
Usage During Study	0.245	0.092	2.652	0.011*	Significant driver

Table 6 – OLS Regression Model 2: Predicting Composite Distraction Score ($R^2 = 0.153$, $F(2,47) = 4.260$, $p = 0.020$ *)

5.5.3 Model 3: Predicting Productivity Impact

The third model examined predictors of perceived productivity impact. While the overall model was not significant ($F(3,46) = 1.807$, $p = 0.159$), SM hours showed a significant negative coefficient ($\beta = -0.407$, $t = -2.171$, $p = 0.035$), indicating that heavier daily users paradoxically perceive less productivity loss — consistent with habituation effects.

Predictor	β	SE	t-stat	p-value	Interpretation
Intercept	2.542	0.534	4.760	< 0.001***	Strong baseline
SM Hours	-0.407	0.188	-2.171	0.035*	Negative association
Focus Reduction	0.134	0.160	0.835	0.408	Non-significant
Procrastination	0.100	0.189	0.530	0.599	Non-significant

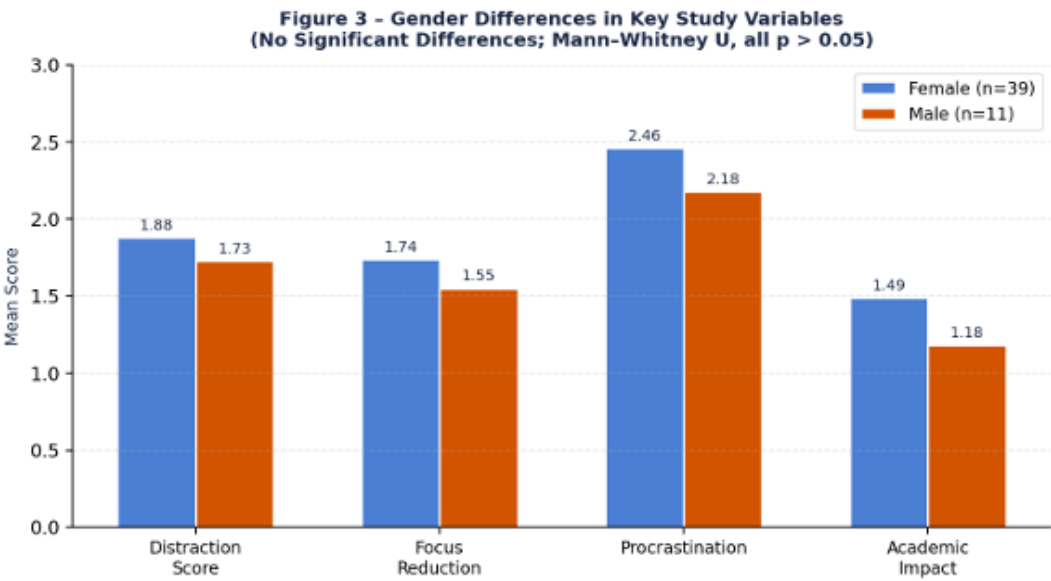
Table 7 – OLS Regression Model 3: Predicting Productivity Impact ($R^2 = 0.105$, $F(3,46) = 1.807$, $p = 0.159$ — Not Significant)

5.6 Gender and Platform Analysis

Research Methods Department Mann–Whitney U tests were conducted for each key outcome variable by gender. Results consistently showed no significant gender differences (Table 8). Females reported marginally higher distraction, focus reduction, and academic impact, but none of these differences approached statistical significance. The negligible effect sizes ($|r_b| < 0.24$) confirm that gender is not a meaningful moderator of social media's academic impact in this sample.

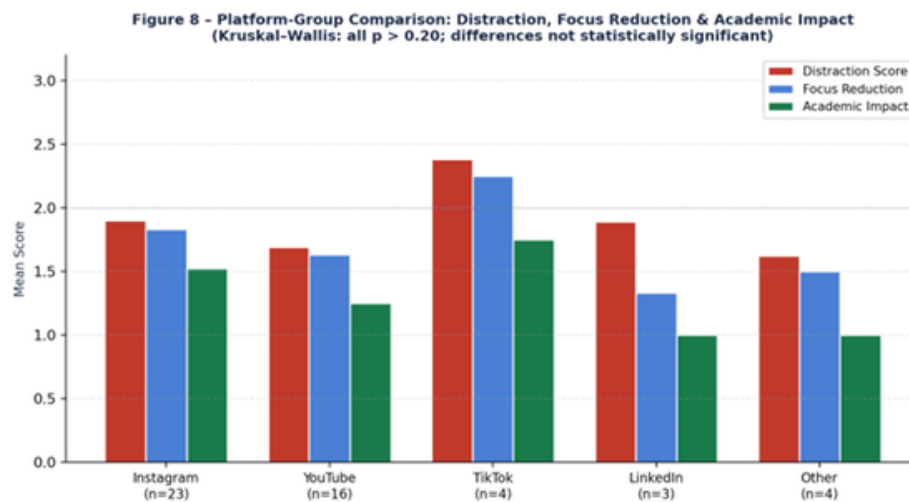
Variable	Female M	Male M	U	p-value	r_b	Decision
Distraction Score	1.880	1.727	248.5	0.430	−0.159	ns
Focus Reduction	1.744	1.545	243.5	0.477	−0.135	ns
Procrastination	2.462	2.182	264.5	0.206	−0.233	ns
Academic Impact	1.487	1.182	251.5	0.355	−0.172	ns

Table 8 – Mann–Whitney U Tests: Gender Differences (Female n = 39, Male n = 11). ns = not significant.



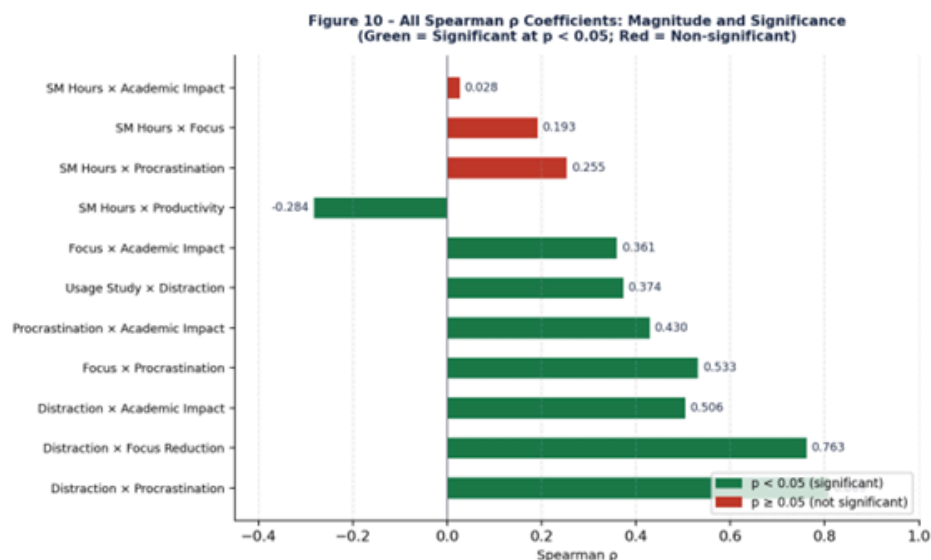
Platform	n	Distraction M (SD)	Academic Impact M (SD)	Focus Reduction M (SD)
Instagram	23	1.90 (0.59)	1.52 (0.73)	1.83 (0.83)
YouTube	16	1.69 (0.67)	1.25 (0.86)	1.63 (0.89)
TikTok	4	2.38 (0.76)	1.75 (0.50)	2.25 (0.50)
LinkedIn	3	1.89 (0.35)	1.00 (0.00)	1.33 (0.58)
Other	4	1.62 (0.25)	1.00 (0.82)	1.50 (0.58)
Kruskal–Wallis	—	H=4.081, p=0.395	H=3.932, p=0.415	H=5.843, p=0.211

Table 9 – Platform-Group Comparisons: Kruskal–Wallis Results



Complete Spearman Correlation Dot-Plot

Figure 10 presents all eleven Spearman correlation coefficients reported in the study as a horizontal bar chart, ranked by absolute magnitude. This visual summary makes immediately apparent the dominant role of the composite Distraction Score in the correlation structure: its associations with procrastination ($\rho = 0.809$) and focus reduction ($\rho = 0.763$) far exceed all other pairwise relationships, confirming the internal validity of the composite as a summary index of social media's academic disruption. The essentially zero correlation between SM hours and academic impact ($\rho = 0.028$, shown in red) provides the clearest evidence that total screen time is a misleading metric for intervention design



Hypthesis Summary

Hypothesis	Decision	Key Statistic	Effect Size
H1: SM Hours $\rightarrow$ Academic Impact	$\times$ Not Supported	$\rho = 0.028$, $p = 0.846$	Negligible
H2: Usage During Study $\rightarrow$ Focus Reduction	$\checkmark$ Supported	$\rho = 0.384$, $p = 0.006^{**}$	Moderate
H3: SM Hours $\rightarrow$ Procrastination	$\sim$ Marginal	$\rho = 0.255$, $p = 0.074$	Small-Moderate
H4: SM Hours $\rightarrow$ Lower Productivity	$\checkmark$ Supported	$\rho = -0.284$, $p = 0.046^*$	Small-Moderate
H5: Gender $\rightarrow$ Distraction Score	$\times$ Not Supported	$U = 248.5$, $p = 0.430$	Negligible
H6: Platform $\rightarrow$ Academic Impact	$\times$ Not Supported	$H = 3.932$, $p = 0.415$	Not computed

Discussion

6.1 Contextual Usage as the Primary Mechanism

The most robust finding of this study is that social media use during study sessions — rather than total daily screen time — is the primary behavioural mechanism through which social media disrupts academic performance. This is evidenced by the significant correlation between usage during study and focus reduction (H2, $\rho = 0.384$), the non-significant correlation between total hours and academic impact (H1, $\rho = 0.028$), and the OLS regression finding that usage during study was the only significant predictor of composite distraction (Model 2, $\beta = 0.245$, $p = 0.011$) even when controlling for total daily hours. These findings align with the attention fragmentation framework (Ophir et al., 2009) and with Rosen et al. (2013)'s experimental demonstration that the primary academic damage occurs during study interruptions rather than from background usage.

6.2 Procrastination as an Academic Impact Driver

Procrastination emerged as the strongest predictor of academic impact in the regression model ($\beta = 0.318$, approaching significance at $p = 0.067$) and showed a significant bivariate correlation with academic impact ($\rho = 0.430$, $p = 0.002$). The very strong correlation between the composite Distraction Score and procrastination ($\rho = 0.809$, $p < 0.001$) suggests that social media's disruptive academic effects are largely mediated through procrastination — students lose track of time, delay starting tasks, and experience study-time deficits that eventually manifest as grade consequences. This supports Andreassen et al. (2016) and Steel's (2007) position on procrastination as the most prevalent form of academic self-regulation failure.

6.3 The Productivity Paradox

A counterintuitive finding was the negative association between daily SM hours and perceived productivity impact (H4, $\rho = -0.284$, $p = 0.046$). The OLS regression confirmed this: heavier daily users rated social media as having less impact on their productivity ($\beta = -0.407$, $p = 0.035$). This "productivity paradox" is consistent with habituation effects — students who have been heavy users for longer periods may have normalised their usage and adapted their self-assessments accordingly. Alternatively, heavy users may engage in motivated reasoning, underestimating the impact of their usage to avoid cognitive dissonance.

6.4 Absence of Gender and Platform Effects The absence of significant gender differences in distraction or academic impact is notable given robust gender differences in social media use documented in the broader literature. One possible explanation is that while females and males differ in platform preferences and usage styles, the academic consequences are similarly distributed. The absence of platform effects (H6) supports the conclusion that behaviour within platforms — checking frequency, passive scrolling, multitasking — rather than platform type per se, drives academic consequences. A student checking Instagram briefly and deliberately between study sessions may be less disrupted than one who passively scrolls YouTube for extended periods.

6.5 Limitations

Several limitations merit acknowledgement. First, the sample ($n = 50$) is modest, reducing statistical power for detecting small effects. The marginal finding for H3 ($p = 0.074$) may reflect a true relationship that remains undetected due to sample size. Second, the study relies entirely on self-report measures, which are subject to social desirability bias. Third, the gender imbalance (78% female) limits generalisability of gender-related findings. Fourth, all significant findings are correlational; the cross-sectional design precludes causal inference.

Conclusion

This study examined the relationship between social media usage and academic performance among 50 university students aged 18–24 by applying a combination of non-parametric statistical tests and Ordinary Least Squares (OLS) regression analysis. The results provide important insights into how social media affects students' academic outcomes and productivity behaviours. Rather than the total amount of time spent on social media being the primary issue, the findings suggest that the academic impact depends more strongly on how and when students engage with social media, particularly during study sessions. This highlights that the behavioural context of usage is more significant than simple screen-time measures. One of the most important findings of the study is that social media use during study hours has a direct association with reduced focus and concentration. Students who frequently checked social media while studying reported greater difficulty in maintaining attention, higher levels of distraction, and lower academic efficiency. This supports the argument that interruptions during cognitively demanding tasks are more harmful than overall usage duration. The study found that composite distraction had a strong positive correlation with procrastination ($\rho = 0.809$) and a moderate positive correlation with academic impact ($\rho = 0.506$), indicating that distraction is not a single isolated factor but a multi-dimensional behavioural pattern that significantly influences students' academic lives.

Among the hypotheses tested, two were clearly supported. First, the hypothesis that social media use during study predicts focus reduction (H2) was strongly confirmed, reinforcing the idea that in-session interruptions are a major source of academic inefficiency. Second, the finding that heavier daily users reported lower perceived productivity loss (H4) presents an interesting paradox. This may suggest that students who use social media more frequently become habituated to its presence and may underestimate its negative effects on their productivity. Such normalization of distraction can make self-assessment unreliable and indicates the need for more objective measurement methods in future studies.

The third hypothesis (H3) showed only a marginal trend, suggesting that while there may be a meaningful relationship, the current sample size may not have been large enough to establish strong statistical significance. This indicates the importance of replication using a larger and more diverse sample. In contrast, no significant differences were found across gender or platform type, suggesting that the negative academic effects of social media are relatively universal across these categories. Whether students used Instagram, WhatsApp, YouTube, or other platforms, the disruptive mechanisms appeared to be similar. The OLS regression analysis further strengthened these conclusions by showing that procrastination, focus reduction, and usage during study together explained 20.1% of the variance in academic impact, with the overall model being statistically significant ($p = 0.015$). Although this percentage does not explain all academic performance differences, it clearly demonstrates that behavioural patterns surrounding social media use are meaningful predictors of academic outcomes. Other factors such as motivation, learning environment, mental health, and study habits may account for the remaining variation.

In conclusion, social media itself is not necessarily the direct cause of poor academic performance; rather, the problem lies in unregulated, frequent interruptions during important academic tasks. The findings suggest that improving students' self-regulation and reducing habitual checking behaviour can significantly improve focus, productivity, and academic success. Future research should include longitudinal designs, objective screen-time tracking, and larger gender-balanced samples to better understand causal relationships and long-term effects. Such studies will help develop stronger evidence-based strategies for managing social media use in academic settings.

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Appendix A Survey Questionnaire

Title: The Impact of Social media Usage on Academic Performance and Productivity

Description: This survey studies how social media use affects your academic performance and productivity. Responses are anonymous and used for academic research only.

Section 1: Demographics

Q	Question	Response Options
Q1	What is your age group?	Below 18 / 18–24 / 25–34
Q2	What is your gender?	Male / Female / Other
Q3	How many hours per day do you spend on social media?	Less than 1 hour / 1–3 hours / More than 3 hours / Other
Q4	How often do you use social media while studying or completing academic tasks?	Rarely / Sometimes / Often / Always
Q5	To what extent does social media reduce your ability to concentrate on studying?	Not at all / Slightly / Moderately / Significantly
Q6	How often does social media cause you to procrastinate on assignments or studying?	Rarely / Sometimes / Often / Always
Q7	How often do you check social media while attending classes or studying independently?	Rarely / Sometimes / Often / Always
Q8	To what extent does social media distract you from completing your coursework on time?	Not at all / Slightly / Moderately / Significantly
Q9	To what extent do you struggle with managing study time due to social media usage?	Not at all / Slightly / Moderately / Significantly
Q10	How often do social media notifications interrupt your focus while studying?	Rarely / Sometimes / Often / Always
Q11	How often do you multitask with social media while studying or working on assignments?	Rarely / Sometimes / Often / Always
Q12	How often do you lose track of study time due to social media use?	Rarely / Sometimes / Often / Always
Q13	How often do you delay starting an assignment or studying because of social media?	Rarely / Sometimes / Often / Always
Q14	How often do you try to limit social media usage during work or study hours?	Always / Often / Sometimes / Rarely
Q15	Which social media platform do you use the most?	Facebook / Instagram / LinkedIn / YouTube / Snapchat / X / Other
Q16	How often does social media impact your academic performance?	Always / Often / Sometimes / Rarely / Never
Q17	To what extent do you feel more productive when you avoid social media while studying?	Not at all / Slightly / Moderately / Significantly
Q18	How often do you take breaks from studying to use social media?	Rarely / Sometimes / Often / Always
Q19	Do you feel that social media affects your academic productivity?	Not at all / Slightly / Moderately / Significantly